

Ajay Jagini

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PROFESSIONAL SUMMARY

Data Analyst with 3+ years of experience in predictive analytics, deep learning, and scalable ML solutions. Currently pursuing an M.S. in Data Science (GPA 3.89) with hands-on expertise in Python, GCP (BigQuery, Dataflow, Vertex AI), and TensorFlow. Built and deployed real-time models for urban mobility, crime analytics, and risk forecasting. Certified Google Cloud Data Engineer with a strong focus on LSTM, CNN, model evaluation, and production-grade pipelines.

EDUCATION

Florida State University

Master of Science in Data Science (Computer Science)

Aug 2023- May 2025

CGPA: 3.897/4.0

Kakatiya Institute of Technology & Science

Bachelor of Technology in Computer Science

Aug 2018 - May 2022

CGPA: 9.27/10

WORK EXPERIENCE

Florida State University

Tallahassee, FL

Conference Assistant (University Housing)

May 2024 - Aug 2024

- Supported business operations by managing housing data for over 500+ students using internal dashboards and cloud-based tools.
- Ensured complete accuracy in data entry and reporting by applying basic statistical methods to support operational decision-making.
- Collaborated with university staff to generate weekly occupancy summaries and logistics reports, improving planning efficiency by 25%.
- Utilized Microsoft Excel with pivot tables, Google Sheets with formulas and conditional logic, and internal cloud-based housing systems to automate reporting workflows and enhance data visibility for housing operations.

Spring ML, Inc. (Egen)

Hyderabad, INDIA

Data Engineer

Jul 2022 – Jul 2023

- Spearheaded training initiatives on Twilio, Terraform, and Google Cloud, equipping the team with advanced skills in DevOps practices; resulted in 30% increase in system uptime and streamlined deployment processes.
- Developed virtual chat bots using Dialog Flow CX framework to enhance customer service experience in a client project; achieved 20% increase in customer satisfaction and reduced response time by 50%.
- Led research initiatives concentrating on front-end technologies such as Flutter and Angular for Progressive Web Apps played a key role in optimizing performance and achieving a 30% reduction in load times.
- Implemented highly efficient Dataflow pipelines for a client project, accelerating data processing by 45% and enabling real-time analytics for informed decision-making.

SpringML, Inc. (Egen)

Hyderabad, INDIA

Consultant Data Engineer (Internship)

Jan 2022 – Jun 2022

- Received training sessions on diverse Google Cloud services.
- Acquired certification as an Associate Cloud Engineer, showcasing comprehensive understanding of Google Cloud services; proficient in optimizing infrastructure, managing resources, and implementing robust security measures.
- Collaborated with team members to plan and execute company-wide team-building initiatives, resulting in a 25% increase in employee engagement and a 20% improvement in interdepartmental communication.

Kakatiya Institute of Technology & Science

Hanamkonda, INDIA

Cloud Computing Lab Support Intern

Oct 2021 - May 2022

- Managed cloud-based tools and services to support academic labs, facilitating access for over 200+ students with 99% platform uptime throughout the semester.
- Provided technical assistance across 30+ lab sessions, helping students troubleshoot cloud deployment issues and software usage—leading to a 40% reduction in support tickets.

- Assisted faculty in setting up 10+ virtual lab environments for remote learning, reducing configuration time by 50% through automated scripts and cloud templates.
- Ensured smooth operation of cloud-based educational platforms, contributing to a 95% satisfaction rating in student and faculty feedback surveys.

Internship Studio

Remote, INDIA

Data Analyst Intern

Jun 2020 - Jul 2020

- Built 5+ interactive dashboards using JavaScript, HTML/CSS, and mock data, improving data accessibility for simulated end users by ~40%.
- Analyzed sample datasets (~10,000+ records) to extract patterns and insights, using Excel and JavaScript-based chart libraries (e.g., Chart.js).
- Improved data visualization quality through structured design and clean data transformation, reducing clutter and increasing interpretability by ~30%.
- Designed user-focused reports showcasing trends and key metrics across multiple sectors (e.g., tech news, user traffic).
- Collaborated with a team of 4+ to simulate a real-world data analysis workflow, including data wrangling, presentation, and feedback integration.

Kakatiya Institute of Technology & Science

Hanamkonda, INDIA

Learning Analytics Research Intern

Oct 2019 - Sep 2021

- Conducted performance analytics on over 1,200+ student assessment records to identify learning trends and knowledge gaps, resulting in a 25% improvement in targeted remediation strategies.
- Utilized statistical techniques (e.g., regression analysis, clustering) and machine learning models (e.g., decision trees, logistic regression) to predict student success rates with ~85% accuracy, enabling early intervention for at-risk learners.
- Supported faculty teams in creating adaptive learning plans by delivering insight dashboards and presenting cohort-based performance summaries, which led to a 15% increase in personalized content delivery.
- Assisted in academic research projects involving educational data mining, using tools such as Python (pandas, scikit-learn) and SQL to clean, analyze, and visualize multi-source educational datasets (~50,000 rows).

Kakatiya Institute of Technology & Science

Hanamkonda, INDIA

Academic Data Analyst Intern

Sep 2018 - Oct 2019

- Analyzed student performance data from 1,000+ exam records using Excel, SQL, and Python (Pandas, NumPy), uncovering key performance trends and data quality issues.
- Assisted faculty in evaluating exam results and identifying learning gaps, resulting in a 15% improvement in targeted academic support strategies.
- Designed interactive dashboards in Power BI to visualize academic trends across 10+ courses, reducing manual reporting time by 60%.
- Provided actionable recommendations for curriculum improvements, contributing to a 10% increase in average student scores in follow-up assessments.

PROJECTS

Optimizing Bike-Sharing Station Placement with Deep Learning & Urban Intelligence

Dec 2024

- Developed a deep learning framework (CNN + LSTM) to forecast demand spikes across metro zones—techniques adaptable to modeling interest rate fluctuations or treasury liquidity stress zones.
- Aggregated multi-source datasets (usage logs, weather, POIs), mimicking real-world treasury inputs like economic indicators, asset flows, and cash positions.
- Accelerated model execution by up to 86% through parallel processing and data optimization, enabling real-time decision-making for city-scale deployments.
- Delivered actionable insights with top-10 ranked station placement maps, equipping urban planners with scalable, data-driven strategies for sustainable mobility.

Crime Data Analytics & Visualization Using Microsoft Copilot Studio, Power BI

Nov 2024

- Built interactive dashboards with adaptive workflows to track and visualize high-volume incident data—framework applicable for risk heatmaps, liquidity thresholds, and stress indicators.
- Integrated Power BI to handle large financial datasets (>100MB), achieving 60% faster load time and responsive insights, useful for treasury reporting and risk surveillance.

- Developed adaptive cards and query-driven workflows to enable real-time, interactive visualizations including heatmaps, trend lines, and region-wise breakdowns.
- Integrated Power BI to handle large datasets exceeding 100MB, achieving 60% faster load times and improving performance over Flask-based APIs.
- Delivered actionable insights for law enforcement and public policy through dynamic dashboards, streamlining crime trend discovery and hotspot identification across 10+ city regions.

Multivariate Time Series Forecasting For Drought

Oct 2023

- Built univariate and multivariate LSTM models to simulate drought-level progression, applying time-series principles transferable to financial risk metrics such as cash flow forecasting or interest rate scenarios.
- Employed OLS regression and statistical inference to identify temporal patterns—analogueous to liquidity risk or macroeconomic trend modeling in treasury environments.

Diabetes Prediction using ANN

Aug 2022

- Developed an Artificial Neural Network (ANN)-based machine learning model to predict Diabetes, achieving an accuracy rate of 86% on a PIMA dataset.

Image based Steganography using RSA

May 2022

- Designed and implemented a robust data security pipeline integrating RSA encryption and steganography to protect sensitive information, ensuring secure transmission and high data integrity.
- Transformed and preprocessed structured/unstructured data into binary formats for seamless embedding and encryption across text and image types, validated through pixel-level quantitative analysis.
- Developed a modular system simulating real-world secure data exchange with minimal data loss, demonstrating strong problem-solving, cryptographic knowledge, and technical documentation skills.

Automated No Due Certificate Issue Management System

Sep 2021

- Engineered an intuitive interface for efficiently issuing No Due Certificates, resulting in a 60% reduction in processing time and a significant increase in productivity.
- Enhanced student portal capabilities, allowing students to effortlessly monitor due and no-due statuses in various departments; increased operational efficiency by saving an average of 25 minutes per student per inquiry.

News Portal Data Interface

Jul 2020

- Organized and categorized 500+ news articles within a structured web portal, enhancing content discoverability and topic segmentation across 10+ categories.
- Built JavaScript-based search/filter logic enabling real-time dataset access, boosting portal usability by ~50% and simulating live data dashboard features.
- Contributed to content optimization and reporting, assisting in dynamic display updates and creating visual summaries of portal structure and user interaction flows.

ADDITIONAL

Technical Skills:

- **Programming Languages:** Python, SQL, Java, Bash
- **Data Analysis & Visualization:** Pandas, NumPy, Matplotlib, Seaborn, Microsoft Excel (Pivot Tables, Formulas), Power BI, Tableau, Looker Studio, CRM Analytics
- **Quantitative Finance & Risk Analytics:** Time Series Forecasting (Univariate & Multivariate), Cash Flow Modeling, Liquidity & Interest Rate Risk Concepts, Statistical Inference, Treasury Data Analytics, OLS Regression
- **Modeling & Statistical Techniques:** Linear/Logistic Regression, ARIMA, LSTM, CNN, Time Series Analysis, Cross-Validation, Hypothesis Testing, Feature Selection, Model Interpretability (SHAP, Feature Importance), Residual Analysis.
- **Machine Learning & Deep Learning:** Scikit-learn, TensorFlow, Keras, XGBoost, CNN, LSTM, Regression Models (Linear, Logistic, Regularized), Classification, Model Evaluation, Time Series Forecasting, Feature Engineering, Predictive Scoring, Parametric & Non-Parametric Techniques.

- **Cloud Platforms & Services:** Google Cloud Platform (BigQuery, Dataflow, Pub/Sub, Vertex AI, Vertex AI Pipelines, Cloud Storage), Salesforce AI Cloud, Einstein Discovery, Azure (Basics)
- **MLOps & Automation:** MLflow, Vertex AI Pipelines, CI/CD Concepts, Apache Airflow, Terraform, Docker
- **Data Engineering & Big Data Tools:** Apache Spark, PySpark, RESTful APIs, Data Cleaning, Data Wrangling
- **Databases:** BigQuery, MySQL, PostgreSQL, NoSQL (Firestore, MongoDB)
- **Version Control & Operating Systems:** Git, GitHub, Linux/Unix
- **AI & Responsible Practices:** Responsible AI Principles, AI Use Case Evaluation

Certifications & Training: Salesforce AI certified Associate, HashiCorp Terraform Associate (002), Professional Data Engineer, Professional Cloud DevOps Engineer (Google Cloud), Database Design and Programming With SQL (Oracle), Foundations of AI (Wipro).

Awards: Published research paper in IEEE Conference, Secured 75th position out of 17,000 students in Internshala Student Partner (ISP) 14.